Nama: Alia Islami Fastha
NIM : 211331037

No.
Date 28/02/22.

BAB II

2.2 Load-Line Analysis

1. Using the characteristics of Fig. 2.131b. determine I_D, V_D, and V_R for the circuit of Fig. 2.131a.

$\therefore V_{DQ} \cong 0.85\,V$

$I_{DQ} \cong 15\,mA$

$V_R = E - V_{DQ} = 12V - 0.85V = 11.15\,V$

b. Repeat part (a) using the approximate model for the diode and compare result.

$\therefore V_{DQ} \cong 0.7\,V$

$I_{DQ} \cong 15\,mA$

$V_R = E - V_{DQ} = 12\,V - 0.7\,V = 11.3\,V$

c. Repeat part (a) using the ideal model for the diode and compare result.

$\therefore V_{DQ} \cong 0V$

$I_{DQ} \cong 16\,mA$

$V_R = E - V_{DQ} = 12\,V - 0V = 12\,V$

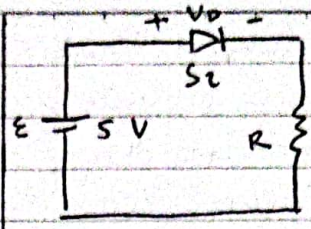
for (a) and (b), Levels of V_{DQ} and I_{DQ} are quite close. Levels of part (c) are reasonably close but as expected close to level of applied voltage E.

2. a. using the characteristics of fig. 2.131b, determine I_D and V_D for the circuit of fig. 2.13.

b. Repeat part (a) with R=0.47kΩ

c. Repeat part (a) with R = 0.18 kΩ

d. Is the level of V_D relatively close to 0.7 V in each case? How do the resulting levels of I_D compare? Comment accordingly.



a. $I_D = \frac{E}{R} = \frac{5}{2.2k\Omega} = 2.27\,mA$

b. $I_D = \frac{E}{R} = \frac{5}{0.47} = 10.6\,mA$

c. $I_D = \frac{E}{R} = \frac{5}{0.18} = 27.7\,mA$

3. Determine the value of R for the circuit of Fig. 2.132 that will result in a diode current of 10 mA if E = 7V use the characteristics of fig. 2.131b for the diode.

$\therefore$ Load Line through $I_{DQ} = 10mA$ of characteristics and $V_D = 7V$ will intersect I_D axis as 11.3 mA

$I_D = 11.3\,mA = \frac{E}{R} = \frac{7V}{R}$

with $R = \frac{7}{11.3\,mA} = 619.47\Omega \cong 0.62\,k\Omega$ Standard resistor.

4. a. Using the approximate characteristic for the Si diode determine the level of V_D, I_D and V_R for the circuit of fig. 2.133

b. Perform the same analysis as part (a) using the ideal model for the diode

c. Do the result obtained in

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parts (a) and (b) suggest that the ideal model can provide a good approximation for the actual response under some conditions?

$\therefore I_D = I_R = \frac{E - V_D}{R}$

$= \frac{30V - 0.7V}{1.5k\Omega} = 19.53\,mA$

$V_D = 0.7,\ V_R = E - V_D = 30 - 0.7$

$= 29.3\,V$

$\therefore I_D = \frac{E - V_D}{R} = \frac{30 - 0}{1.5k\Omega} = 20\,mA$

$V_D = 0V,\ V_R = 30V$

Yes, since $E \gg V_T$ the levels of I_D and V_R are quite close.

5. a. $I = 0\,mA$; diode reverse-biased

b. $V_{20\Omega} = 20V - 0.7V = 19.3V$

(Kirchoff's voltage law)

$I(20\Omega) = \frac{19.3V}{20\Omega} = 0.965\,A$

$V(10\Omega) = 20V - 0.7V = 19.3V$

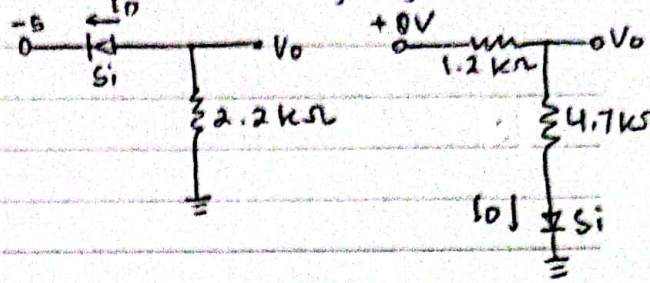
$I(10\Omega) = \frac{19.3V}{10\Omega} = 1.93\,A$

$I = I(10\Omega) + I(20\Omega)$

$= 2.895\,A$

$\therefore I = \frac{10V}{10\Omega} = 1\,A$; center branch open.

6. Determine V_o and I_D for the networks of Fig. 2.135.



a. Dioda forward-biased,

Kirchoff's voltage law (CW)

$-5V - 0.7V - V_o = 0 = -4.3V.$

$I_R = I_D = \frac{|V_o|}{R} = \frac{4.3V}{2.2k\Omega} = 1.955\,mA$

b. Dioda forward-biased,

$I_D = \frac{8V + 6V - 0.7V}{1.2k\Omega + 4.7k\Omega} = 2.25\,mA$

$V_o = 8V - (2.25\,mA)(1.2k\Omega)$

$= 5.3V$

7. Determine the level of V_o for each network of Fig. 2136

$\therefore V_o = \frac{10k\Omega\,(12V - 0.7V - 0.3V)}{2k\Omega + 10k\Omega}$

$= 9.17\,V$

$\therefore V_o = 10V$ //

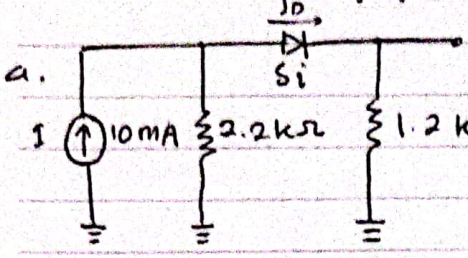
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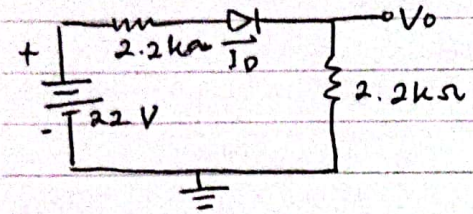
8. Determine V_o and I_D for the networks of fig. 2.137.

a.



$E_{Th} = IR = (10mA)(2.2k\Omega) = 22V$

$R_{Th} = 2.2\ k\Omega$



Diode forward-biased

$I_D = \frac{22V - 0.7V}{2.2\ k\Omega + 2.2\ k\Omega} = 4.84\ mA$

$V_o = I_D(1.2k\Omega)$

$= (4.84\ mA)(1.2\ k\Omega)$

$= 5.81\ V.$

b. Diode forward-biased

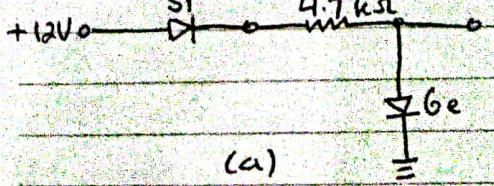
$I_D = \frac{20V + 20V - 0.7V}{6.8\ k\Omega} = 5.78\ mA$

Kirchoff's voltage law (CW)

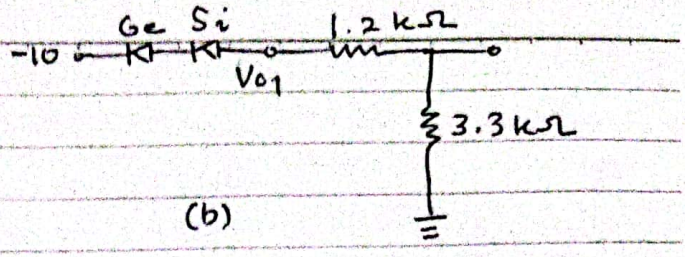
$+V_o - 0.7V + 20V = 0$

$V_o = -19.3\ V$

9. Determine V_{o1} and V_{o2} for the networks of fig. 2.138



(a)



(b)

a. $V_{o1} = 12V - 0.7V = 11.3V$

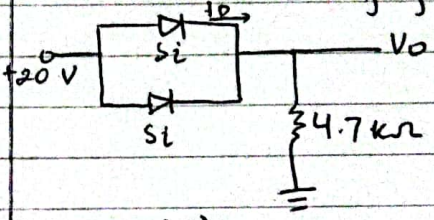
$V_{o2} = 1.2\ V$

b. $V_{o1} = 0V$

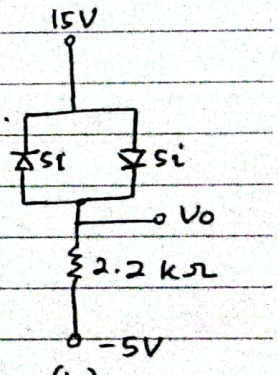
$V_{o2} = 0V$

2.8 Parallel and Series Parallel Configuration

10. Determine V_o and I_D for the networks of fig. 2.139



(a).



(b)

a. Both diodes forward-biased

Si diode turns on first and locks in 0.7 V drop.

$I_R = \frac{12V - 0.7\ V}{4.7\ k\Omega} = 2.mA$

$I_D = I_R = 2.4\ mA$

$V_o = 12V - 0.7V = 11.3\ V$

b. Right diode forward-biased

$I_D = \frac{20V + 4V - 0.7V}{2.2\ k\Omega}$

$= 10.59\ mA$

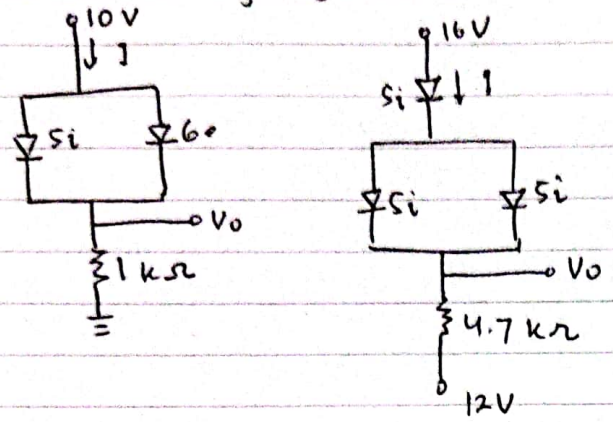
$V_o = 20V - 0.7\ V = 19.3\ V.$

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11. Determine V_o and I for the networks of Fig. 2.140.



a. Si Dioda "on" preventing GaAs diode from turning "on"

$$I = \frac{10V - 0.7V}{1k\Omega} =$$

b. $$I = \frac{16V - 0.7V - 0.7V + 12V}{4.7k\Omega} =$$

$$V_o = 16V - 0.7V - 0.7V = 14.6V$$

12. Determine V_{o1}, V_{o2}, and I for the network of Fig. 2.141

∴ Both diodes forward-biased:

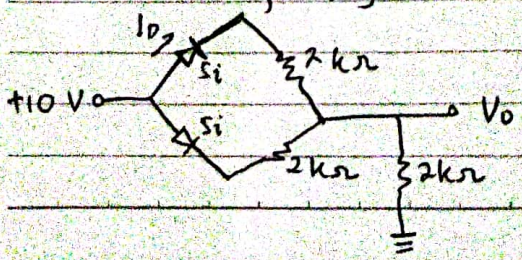
$V_{o1} = 0.7V$, $V_{o2} = 0.7V$

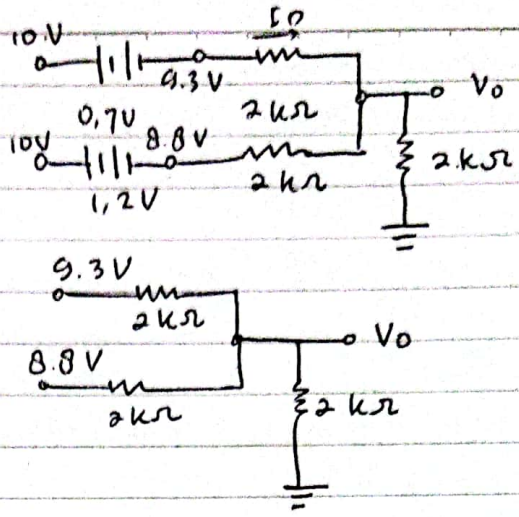
$$I_{1k\Omega} = \frac{20V - 0.7V}{1k\Omega} = \frac{19.3V}{1k\Omega} = 19.3\,mA$$

$$I_{0.47k\Omega} = 0\,mA$$

$$I = I_{1k\Omega} - I_{0.47k\Omega} = 19.3 - 0 = 19.3\,mA$$

13. Determine V_o and I_D for the network of Fig. 2.142.





Superposition:

$$V_{o1}(9.3V) = \frac{1k\Omega\,(9.3V)}{1k\Omega + 2k\Omega} = 3.1V$$

$$V_{o2}(8.8V) = \frac{16k\Omega\,(8.8V)}{7k\Omega + 2k\Omega} = 2.93V$$

$$V_o = V_{o1} + V_{o2} = 6.03V$$

$$I_D = \frac{9.3V - 6.03V}{2k\Omega} = 1.635\,mA$$

2.6 AND/OR Gates

14. Determine V_o for the networks of fig. 2.38 with 0V on both inputs.

∴ Both diodes "off". The threshold voltage of 0.7V is unavailable for either diode

$V_o = 0V$.

15. Determine V_o for the network of Fig. 2.38 with 10 V on both inputs.

∴ Both Diodes "On", $V_o = 10 - 0.7 = 9.3V$.

16. Determine V_o for the network of Fig. 2.41 with 0.V on both inputs.

∴ Both diodes "on".

$V_o = 0.7V$.

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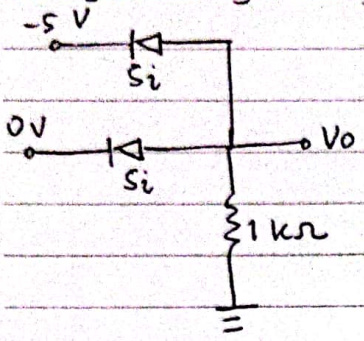
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17. Determine V_o for the network of Fig. 2.41 with 10 V on both inputs.

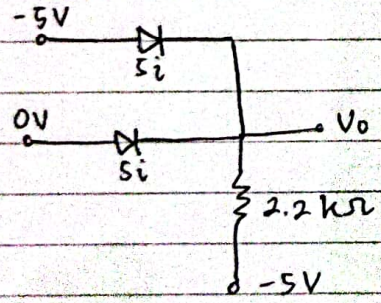
∴ Both diodes "off", $V_o = 10V$

18. Determine V_o for the negative logic OR gate of Fig. 2.143.



The Si diode with −5 V at the ~~that~~ cathode is "on" while the other is "off". The result is $V_o = -5V + 0.7 = -4.3\,V$

19. Determine V_o for the negative logic AND gate of fig. 2.144



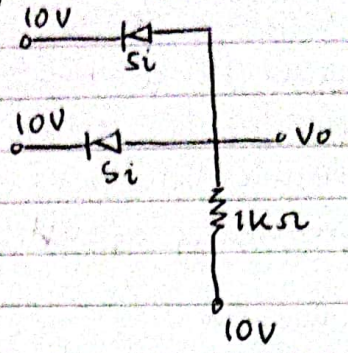
0 volt at one terminal is "more positive" than −5V at the other input terminal. Therefore assume lower diode "on" and upper diode "off"

The result:

$V_o = 0V - 0.7\,V = -0.7V$

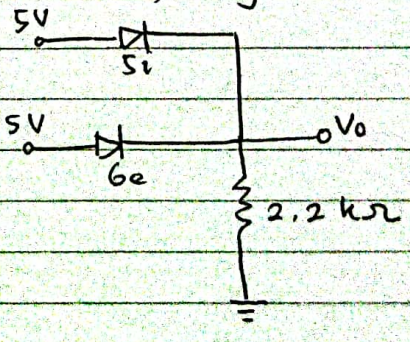
The result supports the above assumptions.

20. Determine the level of V_o for the gate of Fig. 2.145.



Since all the system terminals are at 10 V the required difference of 0.7V accross either diode cannot be established. Therefore, both diodes are "off" and $V_o = +10V$ as established by 10V supply connected to 1 kΩ resistor.

21. Determine V_o for the configuration of Fig. 2.146



∴ The Si Diode requires more terminal voltage than the Ge diode turn "on". Therefore, with 5V at both input terminals, assume Si diode "off" and Ge diode "on".

The result: $V_o = 5V - 0.3V = 4.7V$

The result supports the above assumptions.

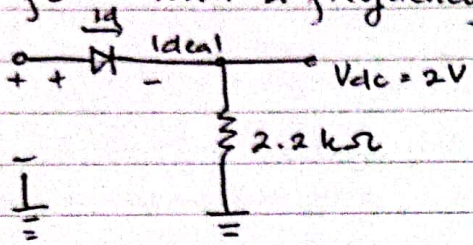
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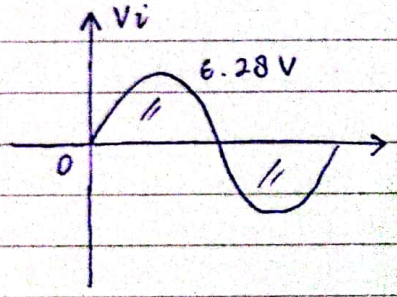
2.7 Sinusoidal Inputs; Half-wave Rectification

22. Assuming an ideal diode, sketch V_i, V_d, and I_d for the half-wave rectifier of Fig. 2.147 The input is a sinusoidal waveform with a frequency of 60 Hz.

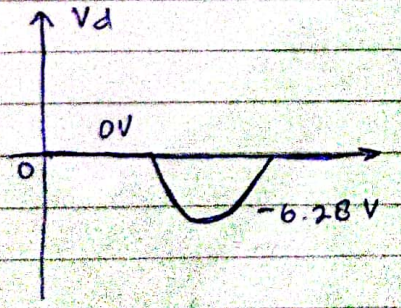


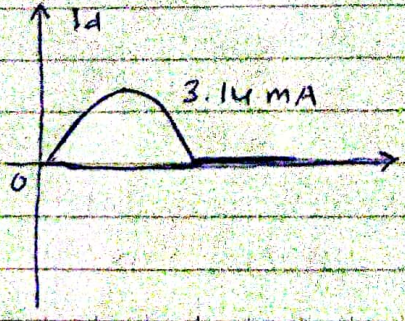
$V_{dc} = 0.318\, V_m \Rightarrow V_m = \frac{V_{dc}}{0.318} = \frac{2}{0.318}$

$= 6.28\ V$



$I_m = \frac{V_m}{R} = \frac{6.28\ V}{2\ k\Omega} = 3.14\ mA$



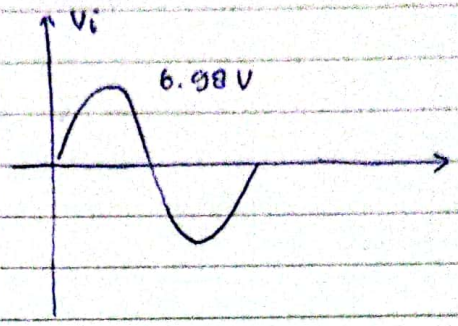


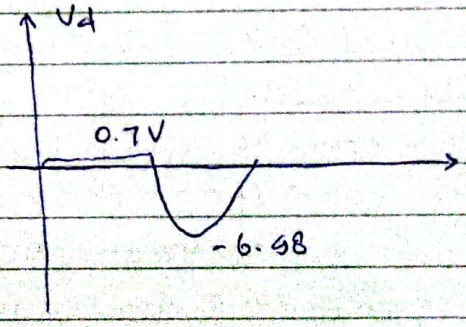
23. Repeat Problem 22 with a silicon diode ($V_T = 0.7\ V$).

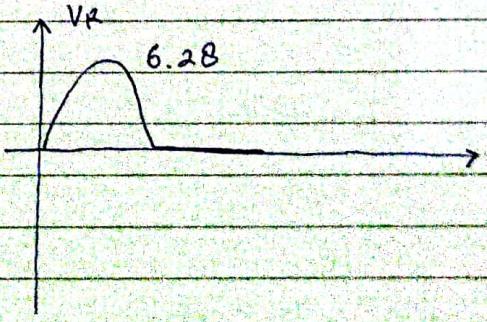
$\therefore$ Using $V_{dc} = 0.318\,(V_m - V_T)$

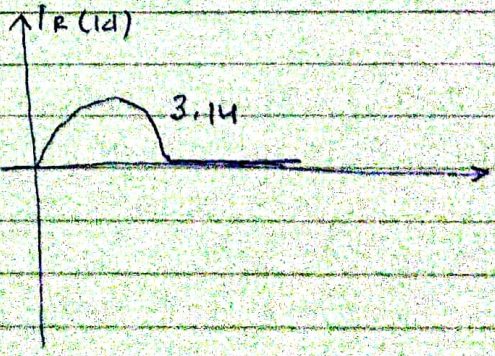
$2V = 0.318\,(V_m - 0.7V)$

Solving $V_m = 6.98\ V \cong 10{:}1$ for $V_m{:}V_T$.







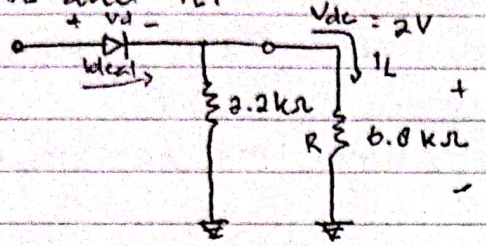


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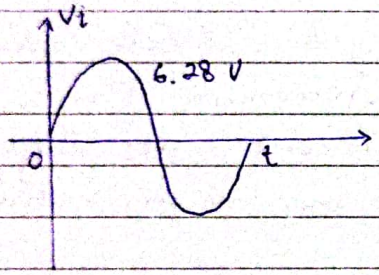
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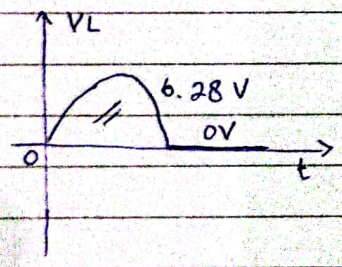
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24. Repeat problem 22 with a 6.8 kΩ load applied as shown in fig. 2.148. Sketch V_L and i_L.

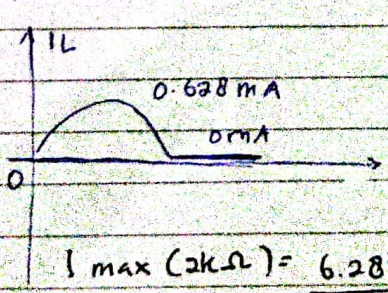


$$V_m = \frac{V_{dc}}{0.318} = \frac{2\ V}{0.318} = 6.28\ V$$



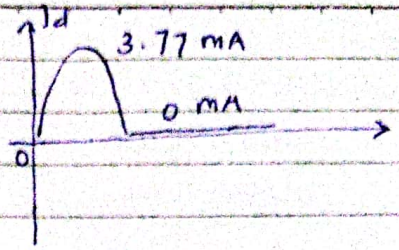


$$I_{L\,max} = \frac{6.28\ V}{10\ k\Omega} = 0.628\ mA$$

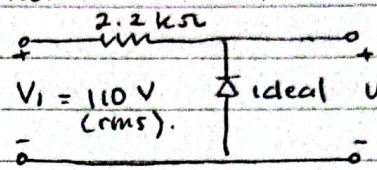


$$I_{max}\,(2k\Omega) = \frac{6.28V}{2k\Omega} = 3.14mA$$

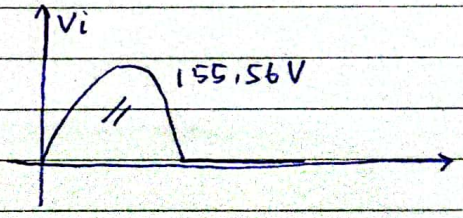
$$I_{D\,max} = I_{L\,max} + I_{max}\,(2k\Omega)$$
$$= 0.678\ mA + 3.14\ mA$$
$$= 3.77\ mA$$



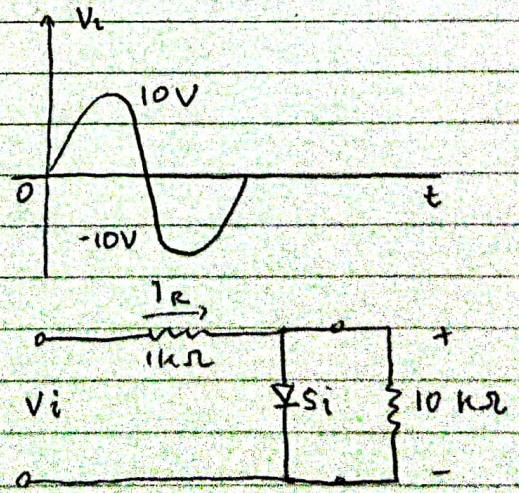
25. For the network of Fig. 2149, Sketch V_o and determine V_{dc}.



$$V_m = \sqrt{2}\,(110V) = 155.56$$
$$V_{dc} = 0.318\ V_m = 0.318\,(155.56)$$
$$= 49.46\ V$$



26. For the network of Fig. 2.150, Sketch V_o and I_R



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Diode will conduct when $V_o = 0.7V$

that is, $V_o = 0.7V = \frac{1k\Omega(V_i)}{1k\Omega + 1k\Omega}$

Solving : $V_i = 1.4V$.

for $V_i \geq 1.4V$ Si diode is "on" and $V_o = 0.7V$

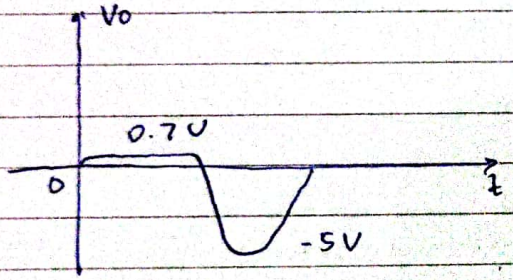
for $V_i < 1.4V$ Si diode is open and level of V_o is determined by voltage divider rule:

$V_o = \frac{1k\Omega}{1k\Omega + 1k\Omega} = 0.5\,V_i$

for $V_i = -10V$

$V_o = 0.5(-10V)$

$= -5$

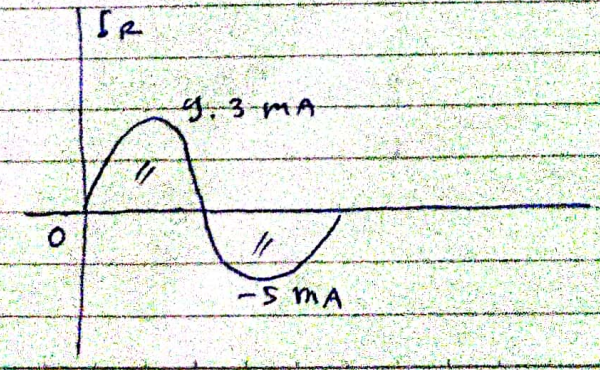


When $V_o = 0.7V$, $V_{R_{max}} = V_{imax}$

$= 10V - 0.7V = 9.3V$

$I_{R_{max}} = \frac{9.3V}{1k\Omega} = 9.3\,mA$

$I_{max\,(reverse)} = \frac{10V}{1k\Omega + 1k\Omega} = 0.5\,mA$

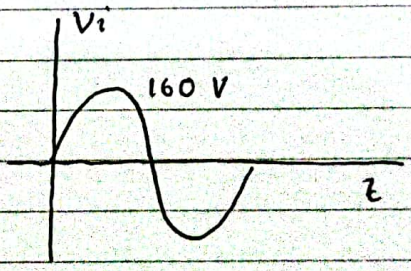


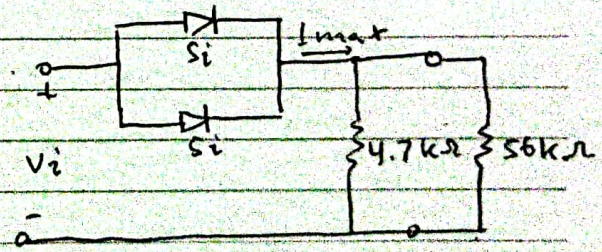
27. a. Given $P_{max} = 14mW$ for each diode of Fig. 2.151 determine the maximum current rating of each diode (using the approximate equivalent model).

b. Determine I_{max} for $V_{max} = 160V$

c. Determine the current through each diode at V_{imax} using the result of part (b).

d. If only one diode were present, determine the diode current and compare it to the maximum rating.





a. $P_{max} = 14mW = (0.7V)I_d$

$I_D = \frac{14mW}{0.7V} = 2mA$

b. $I_{max} = 2 \cdot 20mA = 40mA$

c. $4.7k\Omega \parallel 56k\Omega = 4.3k\Omega$

$V_R = 160V - 0.7V = 159.3V$

$I_{max} = \frac{159.3V}{4.3} = 37.04\,MA$

$I_d = \frac{I_{max}}{2} = 18.1\,mA$

d. Total damage, $37.04 > 20\,mA$

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No.

Date

2.8 Full Wave Rectification

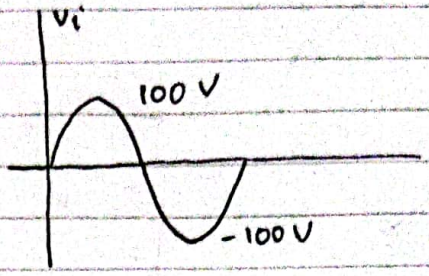
28. A full wave bridge rectifier with a 120-V rms sinusoidal input has a load resistor of 1 kΩ.

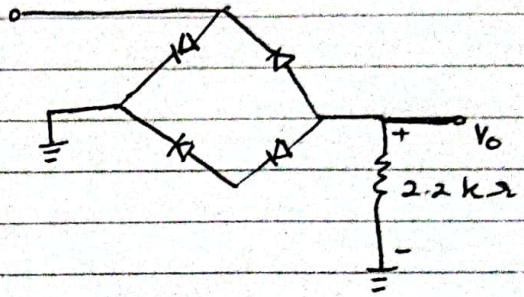
a. If silicon diodes are employed, what is the dc voltage available at the load?

b. Determine the required PIV rating of each diode

c. Find the maximum current through each diode during conduction

d. What is the required power rating of each diode?





a. $V_m = \sqrt{2}(120) = 169.7\ V$

$V_{Lm} = V_{lm} - 2V_D$

$= 169.7V - 2(0.7V)$

$= 169.7 - 1.4$

$= 168.3\ V$

$V_{dc} = 0.636(168.3V) = 107.04V$

b. $PIV = V_m(Load) + V_D$

$= 168.3 + 0.7$

$= 169\ V$

c. $I_{D(max)} = \frac{V_{Lm}}{R_L}$

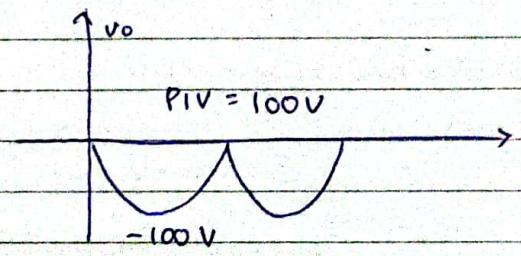
$= \frac{168.3\ V}{1k\Omega} = 168.3mA$

d. $P_{max} = V_D I_D = (0.7V) I_{max}$

$= (0.7V)(168.3mA)$

$= 117.81\ mW.$

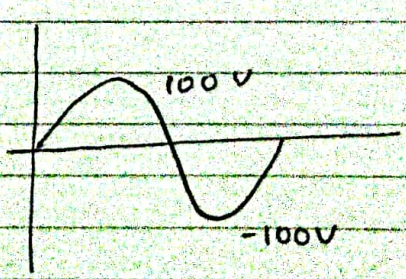
∴



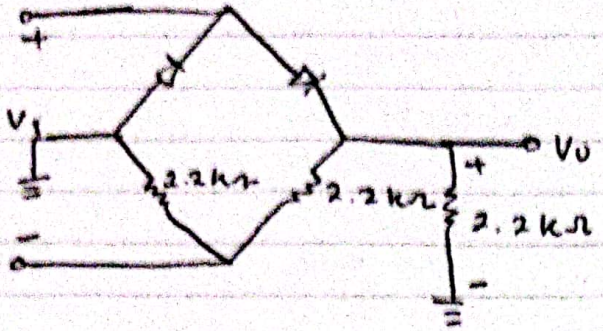
$I_{max} = \frac{100\ V}{2.2k\Omega} = 45.45\ mA$

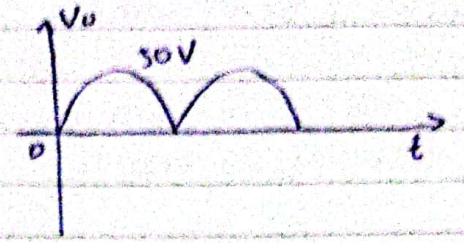
29. Determine V_o and the required PIV rating of each diode for the configuration of fig. 2.152.

30. Sketch V_o for the network of fig. 2.153 and determine the dc voltage available.



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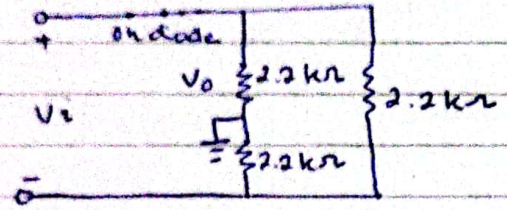
$V_{dc} = 0.636\,V_m$

$= 0.636\,(50\,V)$

$= 31.8\,V$ //

∴ Positive half-cycle of Vi

network redrawn:



Voltage-divider rule;

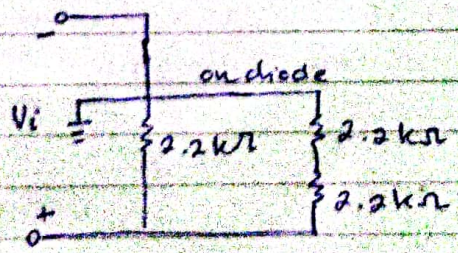
$$V_{omax} = \frac{2.2\,k\Omega\,(V_{imax})}{2.2k\Omega + 2.2k\Omega}$$

$$= \frac{1}{2}\,(V_{imax})$$

$$= \frac{1}{2}\,(100\,V)$$

$$= 50\,V.$$

Negative half-cycle of Vi



Polarity of Vo across the 2.2 kΩ resistor acting as a load is the same.

Voltage-divider rule;

$$V_{omax} = \frac{2.2\,(V_{imax})}{2.2 + 2.2} = \frac{1}{2}\,(V_{imax})$$

$$= \frac{1}{2}\,(100\,V$$

$$= 50\,V$$

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